

MUGE 3D Simulation Entity Framework: GPU Rendering, Per-System Archives, OpenGL

Daniel T. Murphy

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PAPER_168 — MUGE 3D Simulation Entity Framework: GPU Rendering, Per-System Archives, OpenGL

Author: Daniel T. Murphy

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{\text{UQFF}}(r) = g_{\text{MUGE}}(r) \cdot \left(1 - [SSq] \cdot U_{b_i} / F_U(r, t)\right), \quad [SSq] = 0.57$$

Abstract

This paper documents the **MUGE 3D Simulation Entity Framework** — the architectural mapping from MUGE physics systems to 3D renderable entities in the Star Magic visualization engine. Each MUGE system (SGR 1745, SgrA*, Tapestry, Westerlund 2, Pillars, Rings, Student’s Guide) is assigned a per-system archive directory containing image assets, video clips, and per-system physics plugin DLLs. The rendering system uses OpenGL/GLFW with multi-viewport Camera, procedural terrain generation (Perlin noise + extrusion + boolean operations), and MicroTeX LaTeX math overlay. This provides the Tier 3 VR/VM 3D gateway for the UQFF physics output from Tier 2 calculations.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Entity Architecture

1.1 MeshData Struct (OBJ Format)

```
struct MeshData {  
    std::vector<glm::vec3>    vertices;    // 3D positions
```

```

    std::vector<glm::vec3>    normals;    // Surface normals (physics shading)
    std::vector<glm::vec2>    texCoords;  // UV coordinates
    std::vector<unsigned int> indices;    // Triangle indices
};

```

1.2 SimulationEntity

```

struct SimulationEntity {
    std::string name;                // System name (e.g., "SGR1745")
    std::string archive_path;        // "archive/{name}/"
    std::string image_path;         // "archive/{name}/image.jpg"
    std::string video_path;         // "archive/{name}/video.mp4"
    std::string plugin_path;        // "archive/{name}/plugin.dll"
    MugecSystem* muge_system;        // Pointer to physics system
    MeshData    mesh;               // 3D geometry
    glm::vec3    position;           // World position
    glm::quat    orientation;        // Orientation quaternion
    float        scale;              // Scale factor
};

```

2. populate_simulation_entities()

```

void populate_simulation_entities(
    std::vector<MugecSystem>& muge_systems,
    std::vector<SimulationEntity>& entities
) {
    for (auto& sys : muge_systems) {
        SimulationEntity ent;
        ent.name          = sys.name;
        ent.archive_path  = "archive/" + sys.name + "/";
        ent.image_path    = ent.archive_path + "image.jpg";
        ent.video_path    = ent.archive_path + "video.mp4";
        ent.plugin_path   = ent.archive_path + "plugin.dll";
        ent.muge_system   = &sys;
        ent.position      = compute_entity_position(sys);
        ent.orientation    = glm::identity<glm::quat>();
        ent.scale         = compute_entity_scale(sys);
        // Generate placeholder mesh
        ent.mesh          = generateSphereMesh(1.0f, 32, 32);
        entities.push_back(std::move(ent));
    }
}

```

3. Per-System Archive Structure

archive/

```

+- SGR1745/
|   +- image.jpg           ? Chandra X-ray image of SGR 1745-2900
|   +- video.mp4           ? Time-lapse magnetar flare simulation
|   +- plugin.dll          ? SGR1745 UQFF physics plugin
+- SgrA/
|   +- image.jpg           ? EHT Sgr A* shadow image (2022)
|   +- video.mp4           ? MUGE resonance field animation
|   +- plugin.dll          ? SgrA* UQFF physics plugin
+- Tapestry/
|   +- image.jpg           ? HST Tapestry nebula image
|   +- video.mp4           ? Star formation field simulation
|   +- plugin.dll
+- Westerlund2/
+- Pillars/
+- Rings/
+- StudentGuide/

```

4. OpenGL Rendering Pipeline

4.1 Multi-Viewport Camera System

```

struct Camera {
    glm::vec3 position;
    glm::vec3 target;
    glm::vec3 up;
    float     fov;           // Field of view [degrees]
    float     near_plane;    // Near clipping
    float     far_plane;     // Far clipping (set to 1e25 for cosmic scale)

    glm::mat4 getViewMatrix() const;
    glm::mat4 getProjectionMatrix(float aspect) const;
};

```

Multi-viewport layout: physics view (left) + simulation view (right) + data HUD (top).

4.2 Entity Scale Calculation

Systems span 13 orders of magnitude in size:

System	Physical Size	Entity Scale Factor
SGR 1745	~10 km	0.001
SgrA*	~12 × Rs	1.0 (reference)
Tapestry	~100 ly	100.0
Westerlund2	~400 ly	400.0
Pillars	~4 ly	4.0
Rings	~1 Gly	1000.0
Student	Hubble vol.	1e13

5. Procedural Terrain Generationcpp

// Perlin noise heightmap

```

MeshData generateProceduralLandscape(int width, int height, float scale) {
    MeshData mesh;
    for (int z = 0; z < height; z++) {
        for (int x = 0; x < width; x++) {
            float y = perlinNoise(x * scale, z * scale);
            mesh.vertices.push_back({x * scale, y, z * scale});
        }
    }
    computeNormals(mesh);    // Phong shading normals
    computeUVs(mesh);        // UV for physics field texture mapping
    return mesh;
}

// Boolean union (CSG)
MeshData booleanUnion(const MeshData& A, const MeshData& B);

// Extrusion along path
MeshData extrudeMesh(const MeshData& profile, const std::vector<glm::vec3>& path);
$$
\begin{aligned}
& \& \text{Physics fields (MUGE compressed, resonance) are mapped to terrain height via: } \\
& \& \$y_{\text{terrain}} = \log_{10}(|g_{\text{MUGE}}|) \cdot \text{scale} \$ \\
& \& \text{---} \\
& \& \text{\textbf{\# 6. MicroTeX LaTeX Overlay}}
\end{aligned}
$$cpp
// Render physics equations as LaTeX overlay via MicroTeX
void renderLatexOverlay(const std::string& latex_str,
                        float x, float y,
                        float font_size = 16.0f) {
    auto tex_renderer = MicroTeX::create();
    tex_renderer->render(latex_str, x, y, font_size);
}

// Example overlays:
renderLatexOverlay("$g_{\text{comp}} = 1.78 \times 10^{39}$", 0.1, 0.9);
renderLatexOverlay("$g_{\text{res}} = 1.66 \times 10^{45}$", 0.1, 0.85);
renderLatexOverlay("$F_U = -2.06 \times 10^{59}$", 0.1, 0.80);

```

7. UQFF Integration with Rendering

The simulation loop: 1. Physics compute: `compute_compressed_MUGE()` / `compute_resonance_MUGE()`
2. Field normalization: $g \rightarrow \log_{10}(|g|)$ for visual range compression 3. Terrain update: heightmap updated from new MUGE values each frame 4. Particle update: `addUQFFBodyForce()` applied to fluid particles 5. LaTeX HUD: Current MUGE values rendered as math equations in overlay 6. Entity update: position + orientation updated from time-evolved UQFF

8. CP/Architecture Integration

This framework resides in **Tier 3 VR/VM** (source2(HEAD PROGRAM).cpp, ~2,625 lines, GPU-heavy). Connection to CP calculators: - CP1/CP2 compute MUGE values ? stored in `uqff_results.json` - `populate_simulation_entities()` reads `uqff_results.json` ? creates 3D entities - Source2 GUI (Tab for 3D Viewer) ? renders entities via OpenGL pipeline - Per-system plugin DLLs can call CP2/CP3 calculators at runtime

Status: ? Complete | **CP Stage:** Architectural/Tier 3 VR/VM **Supersedes:** N/A (new framework) | **Related:** PAPER_072 (source2 GUI arch), PAPER_157 (Solar System entity params), PAPER_168 connects to CP2/CP3 via `uqff_results.json`

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_U B_i$ jet modulation curves and PAPER_1048 for phonon-corrected $M-\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_U_Bi_i buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **magnetar-field** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_B)(\partial^\mu \phi_B) - V(\phi_B) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_B) = \frac{1}{2}m^2\phi_B^2 + \frac{\lambda}{4!}\phi_B^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_B$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_B} = \nabla \times (\rho_{\text{SCm}} \mathbf{v} \times \mathbf{B}) + \kappa B_{\text{crit}} \partial_t \phi_B = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_B = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(- \exp! \left(- \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.171 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 67, \quad n_{\text{channel}} = 13/26$$

Since $p_{\text{DVP}} = 67$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **103 yr** (field decay quiescence):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}} / \rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.171	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 67$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{m} - 2(UQFFvacuumterm) 1.114 \times 10^{-52} \text{m}^{-2}$	Planck 2018	PASS Consistent	
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).

Paper	Title
PAPER_1050	MUGE F_U_Bi_i Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

2 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_s26_coupling.py</code>	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_scm_cross_section.py</code>	Sym-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n/26)$
<code>kozima_wstp_kernel.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_polylog_s26.py</code>	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms

Module	Purpose	Key Result
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{-26}(z) = Li_{-26}(z) = \eta_{-26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * Li_{-26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: $-f_{-26}(q) = \text{Sum}_{\{n=0\}^{\{25\}}} q^{\{n2\}} / (-q;q)_{n^2} - \phi_{-26}(q) = \text{Sum}_{\{n=0\}^{\{25\}}} q^{\{n2\}} / (-q^2;q^2)_n - \psi_{-26}(q) = \text{Sum}_{\{n=1\}^{\{26\}}} q^{\{n2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	8-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \text{Sum } R_n * (1103+26390n) * W_{-26}(n) / C_{-26}$ where $W_{-26}(n) = \text{Prod}_{\{i=1\}^{\{26\}}} [1 + [SSq] \exp(-\kappa_{p,i} * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN_1_CoAnQi.cpp*, and Wolfram kernels (*uqff_kozima_kernel.wl*, *uqff_s26_kernel.wl*, *uqff_mock_theta_pi_kernel.wl*).